

NEURAL NETWORKS

(Important Questions)

Question marked **GREEN** are **MOST IMPORTANT**

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- 1. Differentiate Soft computing and Hard Computing .**
 - 2. What is the necessity of activation function? List the commonly used activation function.**
 - 3. Draw Biological Neural Network and define its parts.**
 - 4. Define Artificial Neural Network. Explain model of artificial neuron along with its characteristics and suitable block diagram.**
 - 5. Explain the various learning techniques with suitable diagram.**
 - 6. Differentiate between Supervised and unsupervised Learning.**

7. What is Bi-directional associative memory(BAM) ? Differentiate between Auto-associative and Hetro- associative memory in detail. Use the Hebb's rule of discrete BAM, find the weight matrix to store the following (binary) input output pattern pairs.

S(1):(1,1,0)

t(1):(1 0)

S(2):(0,1,0)

t(2):(0,1)

8. Explain why Multi-layer feed-forward net is better than single layer net.

9. What do you understand by the term 'training of Neural Network'? Explain all types of training network

10. Write different applications of neural networks and soft computing.

11. How does learning rate play an important role in learning? How can the training of neural network be improved?
12. Explain McCulloch-Pitts model. Implement AND function using this model.
13. Explain various application of artificial neural network.
14. Explain hebb's rule in auto associative neural network.
15. For the given input vectors $S:(S1, S2, S3, S4)$ and output vectors $t:(t1, t2)$, find the weight matrix using hetero associative training algorithm.

S: (S1, S2, S3, S4)

t=(t1, t2)

I- (1 0 1 0)

(1,0)

II- (1 1 0 0)

(1,0)

III (1 1 1 0)

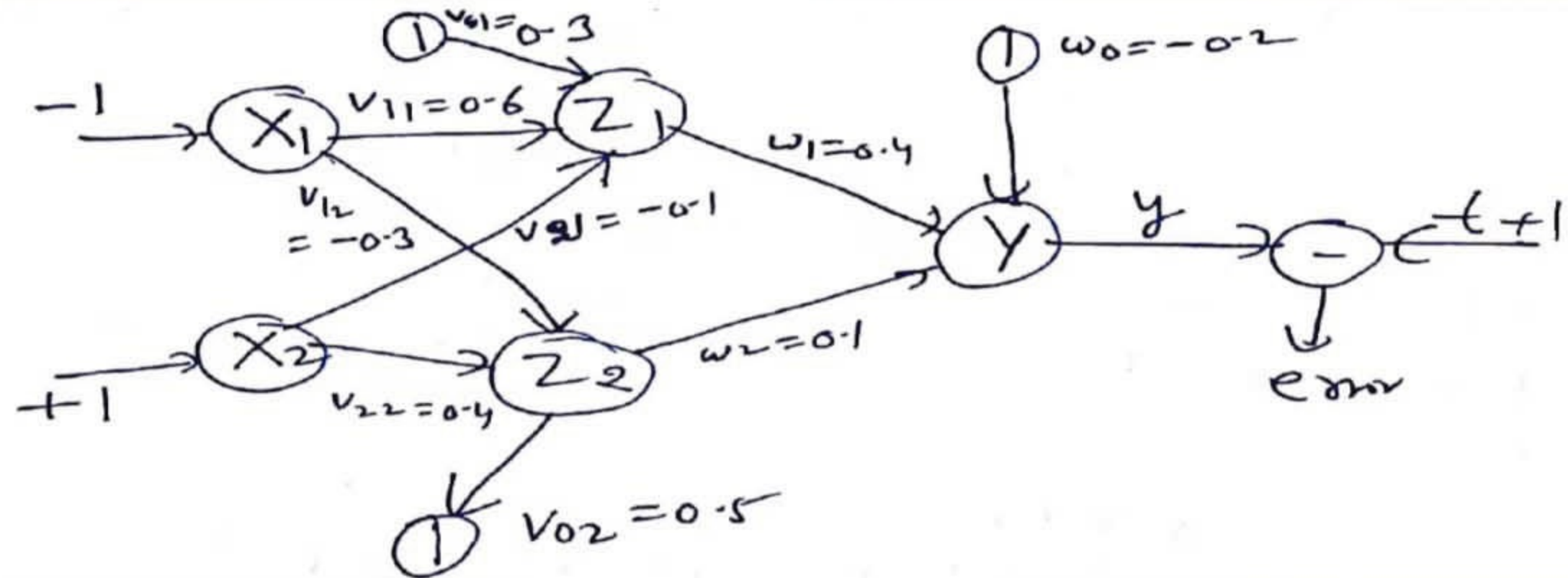
(0,0)

IV- (1 0 0 0)

(0, 1)

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- 16. Explain Back Propagation Network (BPN) and its algorithm with an example.**
 - 17. Explain the following (i)ADALINE (ii) MADALINE. Implement XOR function using MADALINE.**
 - 18. A back-propagation network has 6 neurons in the input layer and 3 neurons in the output layer. What is the no. of neurons in the hidden layer?**
 - 19. Explain the architecture of perceptron neural network model.**
 - 20. Explain the following (i) Delta rule
(ii) Rosenblatt's Perceptron Model**

21. Find the weights required to perform the following classification using perceptron network. The vectors $(1,1,1,1)$ and $(-1,1,-1,-1)$ are belonging to the class (so have target 1), vectors $(1,1,1,-1)$ and $(1,-1,-1,1)$ are not belonging to the class (so have target value -1). Assume learning rate is 1 and weights is 0.
22. Using back propagation network, find the new weights for the network shown in figure. Use a learning rate=0.25 and binary sigmoidal function



23. Discuss the factors affecting the training of Back Propagation Network(BPN).

24. Discuss the selection of various parameter in BPN.

25. Write short notes on

(i) Hopfield network (ii) Adaptive back propagation.

26. Explain self organizing map and its training algorithms.

27. Explain the Rosenblatt's perceptron model with example